Lesson 13.1 - The Least Squares Method

> **Video:** Linear Regression Using Least Squares Method - Line of Best Fit Equation by The Organic Chemistry Tutor **Chapter:** 13 - Regression & Correlation

Goal

Find the straight line that **best fits** a set of points that don't lie exactly on a line. The **least squares method** chooses the line that makes the points as close to it as possible: it minimizes the sum of the squared vertical distances.

The line

Algebra	Statistics	Meaning
$y = mx + b$	$\hat{y} = b_0 + b_1 x$	
m	b_1	slope
b	b_0	y-intercept

Formulas

$$m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - \left(\sum x\right)^2}, \qquad b = \frac{\sum y - m\sum x}{n}$$

n = number of data points. Keep m unrounded when you calculate b.

Example

x	y	xy	x^2
1	1.5	1.5	1
2	3.8	7.6	4
3	6.7	20.1	9
4	9.0	36.0	16
5	11.2	56.0	25
6	13.6	81.6	36

x	y	xy	x^2
7	16.0	112.0	49
Sum	**61.8**	**314.8**	**140**

$\sum x = 28,\ \sum y = 61.8,\ \sum xy = 314.8,\ \sum x^2 = 140,\ n = 7$

Slope:

$$m = \frac{7(314.8) - 28(61.8)}{7(140) - 28^2} = \frac{2203.6 - 1730.4}{980 - 784} = \frac{473.2}{196} \approx 2.414286$$

Intercept:

$$b = \frac{61.8 - 2.414286(28)}{7} = \frac{61.8 - 67.6}{7} = \frac{-5.8}{7} \approx -0.828571$$

Line of best fit:

$$\boxed{y \approx 2.41x - 0.83}$$

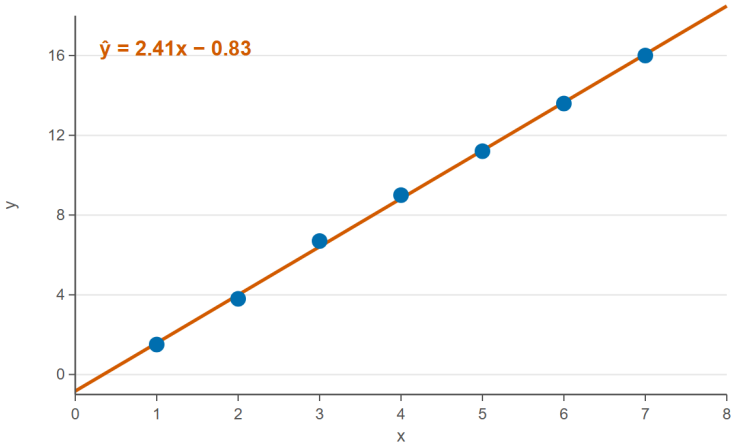
Checking the fit

x	**Predicted** $y = 2.41x - 0.83$	**Actual** y
2	3.99	3.8
5	11.22	11.2
7	16.04	16.0

The predictions are very close to the data, so the line fits well.

Least squares: the line with the smallest sum of squared residuals

Residual = actual y − predicted y (the vertical gaps). The best-fit line makes the sum of their squares as small as possible.



x	y	predicted	residual
1	1.5	1.59	-0.09
2	3.8	4.00	-0.20
3	6.7	6.41	+0.29
4	9	8.83	+0.17
5	11.2	11.24	-0.04
6	13.6	13.66	-0.06
7	16	16.07	-0.07

sum of squared residuals ≈ 0.169

Any other line gives a larger sum.

The residuals are tiny here, so the red gaps on the plot are barely visible.

The least squares line passes very close to every point. The residuals (actual minus predicted) are all small, and their squares add up to about 0.17, which is the smallest possible for any straight line.

Doing it in Excel

With the x values in `B2:B8` and the y values in `C2:C8`:

Want	Formula	Result
Slope m	`=SLOPE(C2:C8, B2:B8)` (known **y**'s first, then known **x**'s)	2.414
Intercept b	`=INTERCEPT(C2:C8, B2:B8)`	−0.8286

Key takeaways

- Build a table with columns x, y, xy, x^2 and add up each column.
- $m = \dfrac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$ and $b = \dfrac{\sum y - m\sum x}{n}$.
- Don't round m before calculating b.
- Excel's `SLOPE` and `INTERCEPT` give the same results in seconds.

Lesson 13.2 - Correlation Coefficient

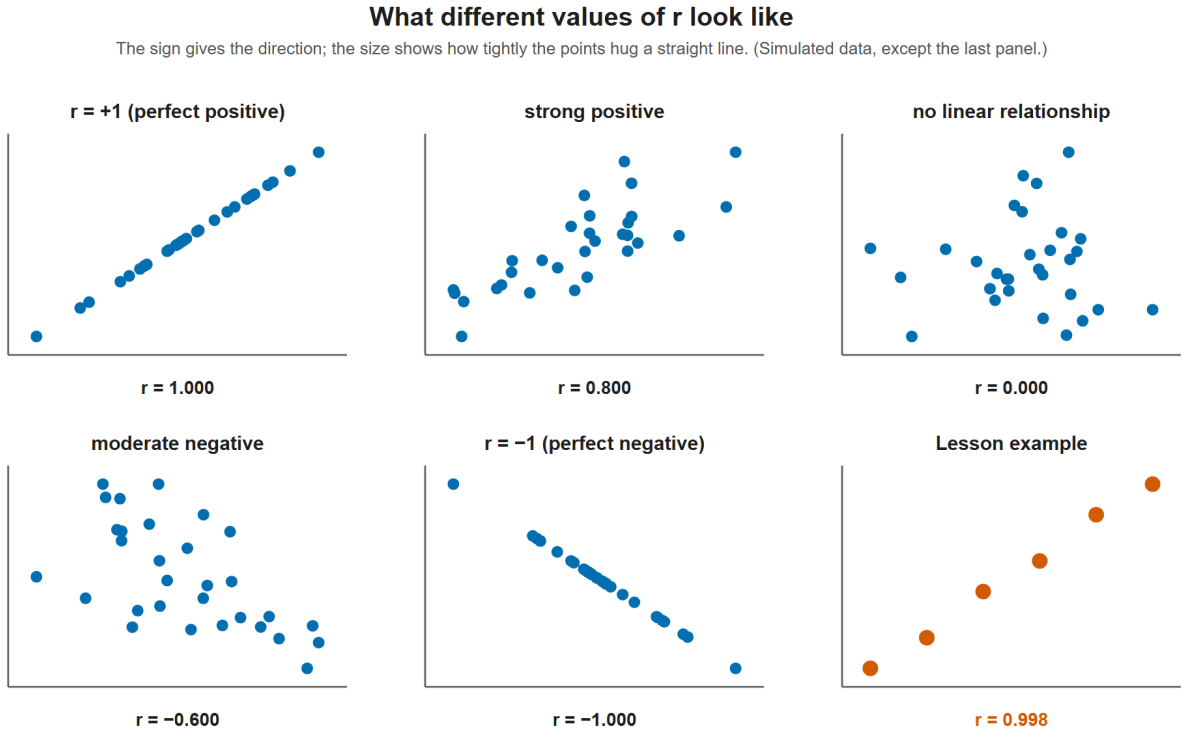
> **Video:** Correlation Coefficient by The Organic Chemistry Tutor **Chapter:** 13 - Regression & Correlation

What r tells you

The **correlation coefficient** r (sometimes written ρ) measures the **strength and direction of the linear relationship** between two variables. $-1 \leq r \leq 1$.

r	Picture	Relationship
$+1$	All points exactly on a line going **up**	Perfect positive (direct): y rises as x rises
-1	All points exactly on a line going **down**	Perfect negative (inverse): y falls as x rises
close to ± 1 (e.g. 0.8)	Points close to a line	Strong linear relationship
smaller (e.g. 0.6-0.7)	Points more scattered around the line	Weaker linear relationship
≈ 0	Random cloud of points	No linear relationship

The **sign** of r matches the sign of the slope. The **size** of r shows how tightly the points hug the line.



The closer r is to Â±1, the more tightly the points line up. The sign follows the direction of the trend. The lesson's data (orange) is almost a perfect line, r = 0.998.

Formula

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left[n\sum x^2 - (\sum x)^2\right]\left[n\sum y^2 - (\sum y)^2\right]}}$$

n = number of (x, y) pairs.

Example

x	y	xy	x^2	y^2
1	2	2	1	4
2	4	8	4	16
3	7	21	9	49
4	9	36	16	81
5	12	60	25	144
6	14	84	36	196
Sum	**48**	**211**	**91**	**490**

$\sum x = 21,\ \sum y = 48,\ \sum xy = 211,\ \sum x^2 = 91,\ \sum y^2 = 490,\ n = 6$

Numerator: $6(211) - 21(48) = 1266 - 1008 = 258$

Denominator:

- $6(91) - 21^2 = 546 - 441 = 105$
- $6(490) - 48^2 = 2940 - 2304 = 636$
- $\sqrt{105 \times 636} = \sqrt{66{,}780} \approx 258.42$

$$r = \frac{258}{258.42} \approx 0.998$$

A **very strong positive** linear relationship: as x increases, y increases almost exactly along a straight line.

Key takeaways

- r ranges from −1 to +1. The sign gives the direction and the size gives the strength of the **linear** relationship.
- Build a table of x, y, xy, x^2, y^2, add each column, then use the formula.
- $r \approx 0$ means no *linear* relationship. There could still be a curved one.
- Correlation is not causation.

Lesson 13.3 - How to Calculate the Covariance

> **Video:** How To Calculate The Covariance Between X and Y by The Organic Chemistry Tutor
> **Chapter:** 13 - Regression & Correlation

Formula

$$\operatorname{cov}(x,y) = \frac{\sum (x-\bar{x})(y-\bar{y})}{n-1} \quad \text{(sample)}$$

For a **population**, divide by N instead of $n-1$.

Steps (table method)

1. Columns: x, y, $x-\bar{x}$, $y-\bar{y}$, $(x-\bar{x})(y-\bar{y})$.
2. Find $\bar{x}$ and $\bar{y}$.
3. Fill in the deviation columns. Each one sums to 0, which is a useful check.
4. Multiply the deviations row by row and add the products.
5. Divide by $n-1$.

Example 1 - positive covariance

x	y	$x-\bar{x}$	$y-\bar{y}$	**product**
2	3	−4	−7.2	28.8
4	7	−2	−3.2	6.4
6	10	0	−0.2	0
8	14	2	3.8	7.6
10	17	4	6.8	27.2
Sum 30	**Sum 51**	0	0	**70**

$\bar{x} = 30/5 = 6,\ \ \bar{y} = 51/5 = 10.2$

$$\operatorname{cov}(x,y) = \frac{70}{5-1} = 17.5$$

Example 2 - negative covariance

x	y	$x - \bar{x}$	$y - \bar{y}$	**product**
3	20	−6	7.4	−44.4
6	17	−3	4.4	−13.2
9	13	0	0.4	0
12	9	3	−3.6	−10.8
15	4	6	−8.6	−51.6
Sum 45	**Sum 63**	0	0	**−120**

$\bar{x} = 45/5 = 9,\ \bar{y} = 63/5 = 12.6$

$$\text{cov}(x, y) = \frac{-120}{5 - 1} = -30$$

(The video has a few slips while adding up this column, such as "−1.8" and "−20", but its final answer, −30, is correct.)

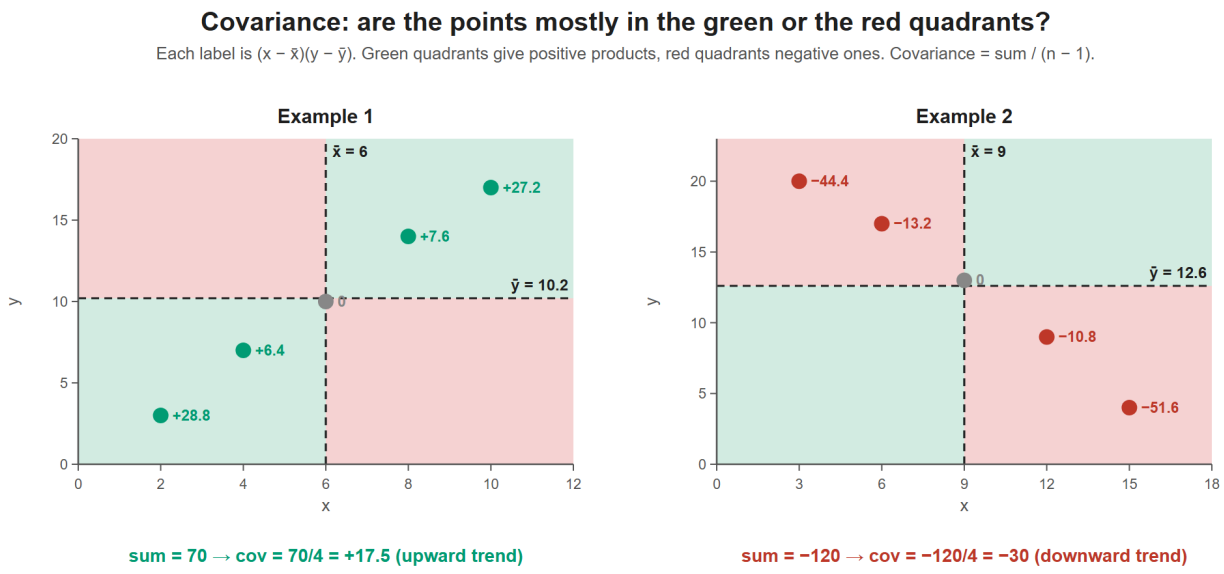
What the sign means

On a scatter plot:

- Example 1: as x increases, y increases (an upward trend) → covariance **positive** (+17.5).
- Example 2: as x increases, y decreases (a downward trend) → covariance **negative** (−30).

Covariance	Relationship
> 0	Positive: the variables move **together**
< 0	Negative: when one goes up, the other goes **down**
≈ 0	No linear relationship

> The **size** of the covariance depends on the units of x and y, so it is hard to say whether 17.5 is "big". The **correlation coefficient** (Lesson 13.4) rescales it to the range −1 to +1.



Points up-right or down-left of the means (green) add positive products. Points up-left or down-right (red) add negative products. Example 1 is all green, so the covariance is positive. Example 2 is all red, so it is negative.

Key takeaways

- Sample covariance $= \sum(x - \bar{x})(y - \bar{y})/(n - 1)$.
- The **sign** gives the direction of the relationship. The size is unit-dependent.
- The deviation columns always add to 0, which is a good check on your arithmetic.

Lesson 13.4 - Calculating the Correlation Coefficient Using the Covariance Formula

> **Video:** How To Calculate The Correlation Coefficient Using The Covariance Formula by The Organic Chemistry Tutor **Chapter:** 13 - Regression & Correlation

Two equivalent formulas

Method 1 - from the covariance:

$$r = \frac{\operatorname{cov}(x,y)}{s_x\, s_y}, \qquad \operatorname{cov}(x,y) = \frac{\sum (x-\bar{x})(y-\bar{y})}{n-1}, \qquad s_x = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}}$$

Dividing the covariance by both standard deviations removes the units and puts r on the −1 to +1 scale.

Method 2 - the computational formula (Lesson 13.2):

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left[n\sum x^2 - (\sum x)^2\right]\left[n\sum y^2 - (\sum y)^2\right]}}$$

Data

x: 3, 4, 6, 8, 9 $\qquad$ y: 4, 5, 9, 10, 12 $\qquad$ $n = 5$

Method 1 - covariance and standard deviations

$\bar{x} = 30/5 = 6,\ \ \bar{y} = 40/5 = 8$

x	y	$x-\bar{x}$	$y-\bar{y}$	**product**	$(x-\bar{x})^2$	$(y-\bar{y})^2$
3	4	−3	−4	12	9	16
4	5	−2	−3	6	4	9
6	9	0	1	0	0	1
8	10	2	2	4	4	4
9	12	3	4	12	9	16
Sum		0	0	**34**	**26**	**46**

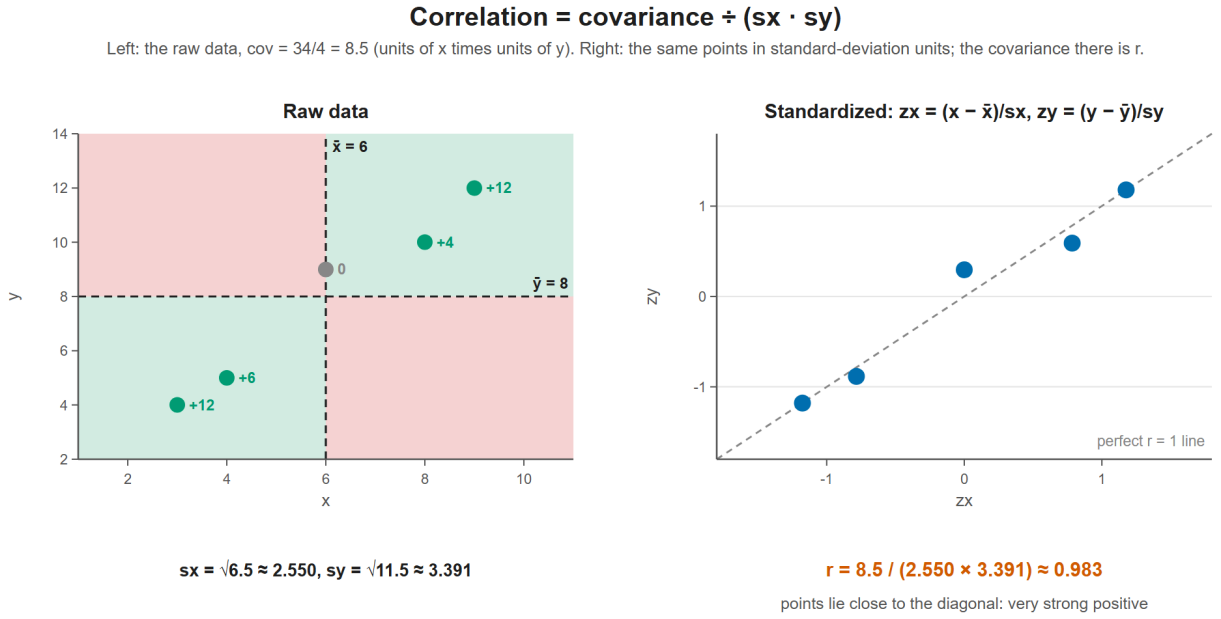
(The video says "pos 2" for the first product at one point. $(-3)(-4) = 12$, and the column total of 34 is correct.)

$$\operatorname{cov}(x, y) = \frac{34}{4} = 8.5$$

$$s_x = \sqrt{\frac{26}{4}} = \sqrt{6.5} \approx 2.54951, \qquad s_y = \sqrt{\frac{46}{4}} = \sqrt{11.5} \approx 3.39116$$

$$r = \frac{8.5}{2.54951 \times 3.39116} \approx 0.9831$$

Mental-math tip: $34/4 = 32/4 + 2/4 = 8 + 0.5 = 8.5$.



Dividing the covariance by sx and sy is the same as measuring both variables in standard-deviation units. In those units the points sit close to the diagonal line zy = zx, which is why r â□□ 0.983 is close to 1.

Method 2 - computational formula (check)

x	y	xy	x^2	y^2
3	4	12	9	16
4	5	20	16	25
6	9	54	36	81
8	10	80	64	100
9	12	108	81	144
30	**40**	**274**	**206**	**366**

- Numerator: $5(274) - 30(40) = 1370 - 1200 = 170$
- $5(206) - 30^2 = 1030 - 900 = 130$
- $5(366) - 40^2 = 1830 - 1600 = 230$

$$r = \frac{170}{\sqrt{130 \times 230}} = \frac{170}{\sqrt{29{,}900}} \approx 0.9831 \checkmark$$

(The product is 29,900; the video says "293,900" at one point.)

Both methods agree: $r \approx 0.983$, a **very strong positive** linear relationship.

Key takeaways

- $r = \dfrac{\operatorname{cov}(x, y)}{s_x s_y}$. Correlation is covariance rescaled to −1 … +1.
- Use $n - 1$ consistently in the covariance and in both standard deviations, or N in all three for a population. The factors cancel either way.
- The computational formula gives the same r without computing deviations.